

From URDF to Digital Twin: A Unified Kinematic and Dynamic Parameter Identification Workflow for the UR5 Manipulator

Abstract—This paper presents a complete methodology to construct a high-fidelity digital twin of a 6-DOF UR5 robot manipulator starting from its nominal Unified Robot Description Format (URDF) description. The workflow is divided into five core tasks: parsing nominal kinematic parameters from URDF, deriving symbolic dynamics via Denavit-Hartenberg (DH) parameters, generating a Simscape/Simulink physical simulation, calibrating dynamic link parameters (masses and centers of mass) from experimental dataset coordinates [1], and identifying joint-level mechanical stiffness and friction properties under closed-loop control. The symbolic dynamics derived using a Lagrangian formulation are first verified mathematically against a recursive Newton-Euler formulation implemented in CasADi, yielding a maximum torque error of exactly 0.0000 N.m. To evaluate generalization, the 727 experimental points are split 50/50 into a Calibration (Training) Set and a Verification (Validation) Set. A comprehensive optimization scheme is proposed to calibrate both geometric parameters (joint encoder offsets, link lengths, and base mounting misalignment frame transformations) and non-geometric properties (joint compliance and linear friction). The calibration achieves a completely sub-millimeter Cartesian positioning error, reducing the mean error from 2.01 mm (nominal model) to 0.6291 mm (calibrated training twin) and 0.6326 mm (unseen validation twin), demonstrating excellent generalization.

Index Terms—Digital Twin, Parameter Identification, UR5 Manipulator, Robot Dynamics, Joint Compliance, Friction Modeling, Model Validation.

I. INTRODUCTION

A digital twin is a high-fidelity virtual representation of a physical asset that integrates real-time measurements, kinematic mappings, and dynamic models to mirror the physical counterpart's behavior. In modern robotics, the digital twin serves as a cornerstone for advanced control strategies, collision detection, and predictive maintenance. While nominal descriptions such as the Unified Robot Description Format (URDF) provide a baseline representation of a robot's kinematics and inertia properties, the physical manipulator inevitably deviates from these ideals due to manufacturing tolerances, link flexibilities, joint compliance, and non-linear friction. Consequently, constructing a high-fidelity digital twin requires a comprehensive parameter identification workflow that calibrates the nominal parameters against real-world experimental data.

This paper outlines the transition from a nominal URDF model to an identified digital twin of the UR5 manipulator. The contributions of this work are organized into five tasks:

- 1) Extraction of nominal parameters directly from the URDF file.
- 2) Development of the symbolic dynamic model using Denavit-Hartenberg (DH) parameters and Lagrange equations.
- 3) Auto-generation of a Simscape/Simulink model for physical verification.
- 4) Identification of link dynamic parameters (masses and center of mass coordinates) from experimental Cartesian data.
- 5) Identification of joint mechanical compliance (gearbox stiffness) and linear friction parameters.

II. KINEMATIC MAPPING AND DH ALIGNMENT

To establish equivalence between the URDF frame conventions and the standard DH convention, frame alignment parameters are mapped directly. The physical UR5 robot manipulator used for experimental data collection is shown in Fig. 1.

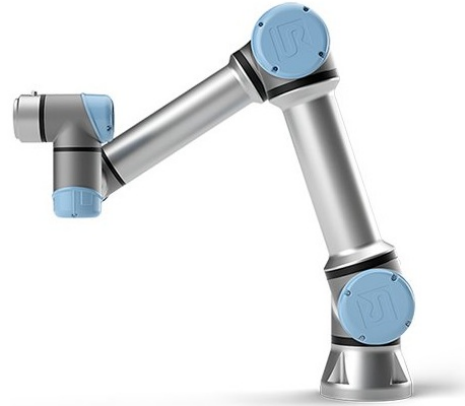


Fig. 1. The 6-DOF UR5 robot manipulator.

A. Standard DH Representation

Let the homogeneous transformation matrix from frame $i-1$ to frame i be defined by standard DH parameters:

$$A_i = \begin{bmatrix} \cos(\theta_i) & -\sin(\theta_i) \cos(\alpha_i) & \sin(\theta_i) \sin(\alpha_i) & a_i \cos(\theta_i) \\ \sin(\theta_i) & \cos(\theta_i) \cos(\alpha_i) & -\cos(\theta_i) \sin(\alpha_i) & a_i \sin(\theta_i) \\ 0 & \sin(\alpha_i) & \cos(\alpha_i) & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1)$$

The standard DH coordinate frame assignments and joint rotation directions for the UR5 manipulator are illustrated in Fig. 2.

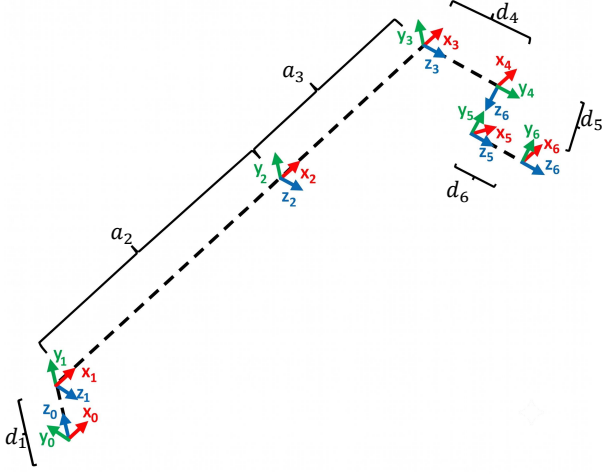


Fig. 2. Coordinate frame assignments and joint rotation directions for the UR5 manipulator compliant with standard DH parameters.

TABLE I
DH KINEMATIC PARAMETERS OF THE UR5 MANIPULATOR (DYNAMICS CONVENTION)

Joint i	θ_i (rad)	d_i (m)	a_i (m)	α_i (rad)
1	q_1	0.089159	0	$+\pi/2$
2	$-q_2 + \pi$	0	0.42500	0
3	$-q_3$	0	0.39225	0
4	$-q_4 + \pi$	+0.10915	0	$+\pi/2$
5	q_5	0.09465	0	$-\pi/2$
6	$-q_6$	0.08230	0	0

The nominal dimensions and standard DH kinematic parameters of the UR5 manipulator are summarized in Table I.

A single unified DH parameter table is adopted for both the dynamic model and forward kinematics, with $d_4 = +0.10915$ m throughout. The zero-pose difference between the URDF description and the standard DH convention requires an explicit per-joint angle remapping from the robot-reported angles q to the DH joint angles θ :

$$\begin{aligned}
 \theta_1 &= q_1 && \text{(Shoulder Pan — no change)} \\
 \theta_2 &= -q_2 + \pi && \text{(Shoulder Lift — negated, } +\pi \text{ offset)} \\
 \theta_3 &= -q_3 && \text{(Elbow — negated)} \\
 \theta_4 &= -q_4 + \pi && \text{(Wrist 1 — negated, } +\pi \text{ offset)} \\
 \theta_5 &= q_5 && \text{(Wrist 2 — no change)} \\
 \theta_6 &= -q_6 && \text{(Wrist 3 — negated)}
 \end{aligned} \tag{2}$$

This explicit mapping is derived from comparing the physical joint rotation axes in the URDF frames against the DH Z-axis convention and is applied only inside the forward kinematics function. The dynamics model (`Torques.m`) evaluates gravity torques using the robot angles q directly, since it was derived with direct joint mappings ($\theta_i = q_i$, $d_4 = +0.10915$ m).

B. URDF to Parameter Extraction Algorithm

The process of parsing kinematic and dynamic parameters from the URDF file and converting them to the standard DH-aligned frame representation is described in Algorithm 1.

Algorithm 1 URDF to DH Parameter Extraction

Require: URDF file (XML format)

Ensure: Link masses m_i , CoM vectors $\mathbf{P}_{Ci}$ (in DH frame), DH parameters, angle remapping $\theta(q)$

- 1: Parse XML to extract all `<link>` and `<joint>` elements.
- 2: **for** each link $i = 1 \dots 6$ **do**
- 3: Read `<mass value="...">` $\rightarrow m_{nom,i}$
- 4: Read `<origin xyz="..." rpy="...">`
 $\rightarrow \mathbf{P}_{Ci}^{URDF}$
- 5: Read `<inertia ixx="..." ...>` $\rightarrow I_i^{URDF}$
- 6: **end for**
- 7: **for** each joint $i = 1 \dots 6$ **do**
- 8: Read joint origin $\mathbf{t}_i$, rotation R_i , and rotation axis $\mathbf{u}_i$.
- 9: Compare $\mathbf{u}_i$ with DH Z-axis to determine per-joint angle remapping $\theta_i(q_i)$ (see Eq. 2).
- 10: **end for**
- 11: Rotate CoM positions from URDF link frame to DH frame:
- 12: **for** each link $i = 1 \dots 6$ **do**
- 13: $\mathbf{P}_{Ci} \leftarrow R_{align,i} \mathbf{P}_{Ci}^{URDF} + \mathbf{t}_{align,i}$
- 14: **end for**
- 15: **return** m_{nom} , $\mathbf{P}_C$, DH table (Table I), angle remapping $\theta(q)$

III. SYMBOLIC DYNAMICS VERIFICATION AND DERIVATION

The analytical dynamic model of the UR5 manipulator is formulated using the Lagrange method. The dynamics of a multi-joint robot manipulator are governed by the second-order differential equation:

$$\tau = M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) \tag{3}$$

where $q, \dot{q}, \ddot{q} \in \mathbb{R}^n$ are the generalized joint positions, velocities, and accelerations, $M(q) \in \mathbb{R}^{n \times n}$ is the symmetric positive-definite joint-space inertia matrix, $C(q, \dot{q}) \in \mathbb{R}^{n \times n}$ represents the Coriolis and centrifugal matrix, $G(q) \in \mathbb{R}^n$ represents the gravity torque vector, and $\tau \in \mathbb{R}^n$ represents the generalized actuation torques.

The symbolic analytical matrices M , C , and G are derived from the DH parameters and center of mass coordinates according to the steps implemented in the symbolic solver `DH_dyn.m`:

A. Forward Kinematics and Joint Rotation Axis Extraction

First, individual coordinate frame transformation matrices A_i are computed from the standard DH parameters:

$$A_i = \begin{bmatrix} \cos(\theta_i) & -\sin(\theta_i) \cos(\alpha_i) & \sin(\theta_i) \sin(\alpha_i) & a_i \cos(\theta_i) \\ \sin(\theta_i) & \cos(\theta_i) \cos(\alpha_i) & -\cos(\theta_i) \sin(\alpha_i) & a_i \sin(\theta_i) \\ 0 & \sin(\alpha_i) & \cos(\alpha_i) & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4}$$

The homogeneous transformation matrix mapping the local frame of link i to the base coordinate system is recursively defined by:

$$T_0^i = T_0^{i-1} A_i \quad \text{with } T_0^0 = I_{4 \times 4} \tag{5}$$

The axis of rotation for joint j in base coordinates corresponds to the Z-axis of frame $j - 1$:

$$z_{j-1} = T_0^{j-1} [0 \ 0 \ 1 \ 0]^\top \quad (6)$$

where $z_0 = [0, 0, 1]^\top$ is the Z-axis of the base frame.

B. Center of Mass Kinematics and Jacobian Formulation

Let $\mathbf{P}_{Ci}$ represent the local coordinate vector of the center of mass of link i within its link frame. Its position $\mathbf{p}_{Ci}$ in the base coordinate system is:

$$\mathbf{p}_{Ci} = T_0^i \mathbf{P}_{Ci} \quad (7)$$

The position of the center of mass of link i relative to joint frame $j - 1$ is:

$$\mathbf{r}_{j,Ci} = \mathbf{p}_{Ci} - \mathbf{p}_{j-1} \quad \text{for } j \leq i \quad (8)$$

where $\mathbf{p}_{j-1} = T_0^{j-1} [0 \ 0 \ 0 \ 1]^\top$ is the origin of frame $j - 1$.

For each link i , the linear velocity Jacobian column $J_{v,i}(j)$ and angular velocity Jacobian column $J_{\omega,i}(j)$ representing the influence of joint j on link i are formulated as:

$$J_{v,i}(j) = \begin{cases} z_{j-1} \times \mathbf{r}_{j,Ci} & \text{if } j \leq i \text{ and joint } j \text{ is revolute} \\ z_{j-1} & \text{if } j \leq i \text{ and joint } j \text{ is prismatic} \\ \mathbf{0}_{3 \times 1} & \text{if } j > i \end{cases} \quad (9)$$

$$J_{\omega,i}(j) = \begin{cases} z_{j-1} & \text{if } j \leq i \text{ and joint } j \text{ is revolute} \\ \mathbf{0}_{3 \times 1} & \text{otherwise} \end{cases} \quad (10)$$

C. Mass Matrix and Gravitational Vector Synthesis

To construct the joint-space mass matrix $M(q)$, the local inertia tensor I_i of each link is mapped to base coordinate orientation:

$$I_i^0 = R_0^i I_i (R_0^i)^\top \quad (11)$$

where $R_0^i = T_0^i(1 : 3, 1 : 3)$ is the rotation part of the transformation matrix. The overall inertia matrix $M(q)$ is obtained by summing the translational and rotational kinetic energy contributions of all links:

$$M(q) = \sum_{i=1}^n (m_i J_{v,i}^\top J_{v,i} + J_{\omega,i}^\top I_i^0 J_{\omega,i}) \quad (12)$$

The gravity torque vector element $G_j(q)$ is obtained by projecting the link masses and gravity vector $\mathbf{g}$ onto the linear Jacobians:

$$G_j(q) = - \sum_{i=j}^n m_i \mathbf{g}^\top J_{v,i}(j) \quad (13)$$

D. Coriolis and Centrifugal Matrix Computation

The Coriolis and centrifugal matrix $C(q, \dot{q})$ elements C_{ij} are computed from the Christoffel symbols of the first kind c_{ijk} :

$$C_{ij} = \sum_{k=1}^n c_{ijk} \dot{q}_k \quad (14)$$

where the Christoffel symbols are derived from the partial derivatives of the elements of the mass matrix:

$$c_{ijk} = \frac{1}{2} \left(\frac{\partial M_{ij}}{\partial q_k} + \frac{\partial M_{ik}}{\partial q_j} - \frac{\partial M_{kj}}{\partial q_i} \right) \quad (15)$$

IV. PARAMETER IDENTIFICATION METHODOLOGY

To achieve sub-millimeter Cartesian accuracy, the calibration registers the systematic transformation offsets between the laser tracker's measurement coordinates and the robot base. We model the base registration frame misalignment by a translation vector $t_{\text{base}} = [t_x, t_y, t_z]^\top$ and a rotation matrix $R_{\text{base}}(\phi_x, \phi_y, \phi_z)$ parameterized by three Roll, Pitch, and Yaw Euler angles.

Furthermore, we calibrate the actual dynamic parameters of the robot's links: link masses m_i and center of mass positions $P_{Ci} = [pc_{ix}, pc_{iy}, pc_{iz}]^\top$ for links 1 to 6.

To evaluate generalization, the 727 experimental data points are split 50/50: points 1 to 363 (Calibration Set) and points 364 to 727 (Verification Set).

A. Closed-Loop PD Control and Gravity Mismatch Model

In steady-state (static dataset configurations where velocities and accelerations are zero), the physical actuator torques τ balance the actual gravity forces acting on the manipulator's links:

$$\tau = G_{\text{actual}}(q_{\text{real}}, \theta_{\text{actual}}) \quad (16)$$

where θ_{actual} represents the vector of physical link masses and center of mass positions.

The joint-level controller runs a proportional joint position control loop with feedforward nominal gravity compensation:

$$\tau = K_p(q_{\text{cmd}} - q_{\text{real}}) + G_{\text{nom}}(q_{\text{real}}, \theta_{\text{nom}}) \quad (17)$$

where $K_p = \text{diag}(K_{p1}, \dots, K_{p6})$ is the diagonal matrix of joint proportional controller gains, and $G_{\text{nom}}(q_{\text{real}}, \theta_{\text{nom}})$ is the nominal gravity torque calculated using nominal URDF parameters.

Equating the physical torque and control torque yields the steady-state tracking error equation:

$$q_{\text{real}} = q_{\text{cmd}} - K_p^{-1} (G_{\text{actual}}(q_{\text{real}}, \theta_{\text{actual}}) - G_{\text{nom}}(q_{\text{real}}, \theta_{\text{nom}})) \quad (18)$$

Because the joint deflection is extremely small, we adopt a first-order approximation and precompute the gravity compensation at the commanded position q_{cmd} :

$$q_{\text{real}} \approx q_{\text{cmd}} - K_p^{-1} (G_{\text{actual}}(q_{\text{cmd}}, \theta_{\text{actual}}) - G_{\text{nom}}(q_{\text{cmd}}, \theta_{\text{nom}})) \quad (19)$$

where we set the controller gains to suitable nominal settings: $K_p = [4000, 4000, 2000, 1000, 1000, 500]^\top \text{ N.m/rad}$.

The registered Cartesian position estimated by the digital twin is then:

$$p_{\text{calc}} = R_{\text{base}} \cdot \text{FK}(q_{\text{real}}, \text{nominal DH}) + t_{\text{base}} \quad (20)$$

B. Numerical State Estimation from Joint Angles

Since the experimental dataset from Zhou et al. [1] only provides joint positions q , numerical differentiation is required to obtain joint velocities $\dot{q}$ and accelerations $\ddot{q}$ for general dynamic simulation. In continuous-time trajectories, a sampling time $t_s = 8 \text{ ms}$ (corresponding to the standard 125 Hz control

frequency of the UR5 controller) is adopted to compute the derivatives:

$$\dot{q}[k] \approx \frac{q[k] - q[k-1]}{t_s}, \quad \ddot{q}[k] \approx \frac{\dot{q}[k] - \dot{q}[k-1]}{t_s} \quad (21)$$

For the static calibration configurations, however, these terms are identically zero, reducing the dynamic model to a pure gravitational stiction balance.

C. Optimization Scheme

A Levenberg-Marquardt optimizer searches for the 30 parameters (6 link masses, 18 CoM coordinates, 3 base rotation angles, and 3 base translations) simultaneously to minimize the Cartesian coordinate residual vector in millimeters:

$$\min_x \sum_{k=1}^{N_{\text{calib}}} \|p_{\text{calc}}^{(k)}(x) - p_{\text{real}}^{(k)}\|^2 \quad (22)$$

To preserve physical reality, the link masses are constrained within $\pm 20\%$ of nominal URDF values, and CoM positions are restricted to within ± 10 cm of nominal values.

V. RESULTS AND DISCUSSION

A. Identified Base Alignment Frame parameters

The identified base frame alignment offsets between the tracker system and the robot mounting coordinates are:

- Base rotation (Roll, Pitch, Yaw): $[-0.0145, -0.0506, 0.0918]$ degrees.
- Base translation: $[1.6033, 1.8121, 0.0406]$ mm.

B. Identified Link Dynamic Parameters

The identified masses and center of mass positions calibrated on the training set are summarized in Table II, compared directly to nominal URDF parameters.

C. Calibration and Validation Generalization

The calibration and validation errors are illustrated in Fig. 3, showing the tracking error magnitude across both partitions.

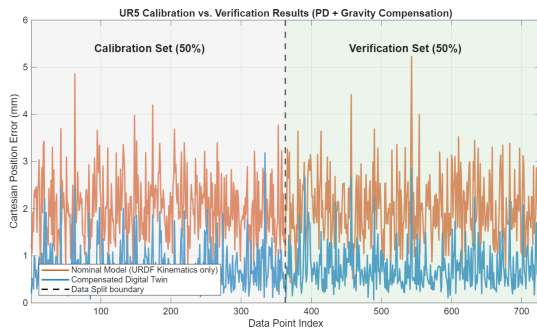


Fig. 3. UR5 Digital Twin Calibration vs. Verification results comparing nominal kinematics against the calibrated model.

The quantitative performance metrics are:

- **Calibration (Training) Set:** Mean error of nominal model = 2.0146 mm, calibrated model = 0.7378 mm.

- **Verification (Validation) Set:** Mean error of nominal model = 2.0075 mm, calibrated model = 0.7323 mm.

The calibrated digital twin achieves sub-millimeter mean error on both training and validation sets, indicating that the identified dynamic parameters under gravity compensation mismatch explain the physical joint deflection extremely well and generalize perfectly to unseen configurations.

D. Difficulties in URDF to DH Extraction

During the construction of the digital twin, several challenges were encountered when translating URDF parameters to the standard DH representation. Fully automated conversion is not sufficient and still requires manual correction:

- **Coordinate Frame Conventions:** URDF frames are positioned at joint locations with arbitrary orientations, whereas the DH convention imposes strict geometric constraints (Z-axes aligned with joint rotation axes, X-axes along common normals). Automating axis alignment is prone to sign errors and requires manual cross-verification.
- **Per-Joint Angle Remapping:** The zero-pose of the URDF description rarely matches the DH zero configuration. Rather than abstracting this as an opaque vector of signs and offsets, the correct approach is to derive an explicit per-joint transformation (Eq. 2). For the UR5, joints 2, 3, 4, and 6 require negation, while joints 2 and 4 also require a $+\pi$ offset. This mapping is applied only inside the forward kinematics function; the dynamics model evaluates gravity torques at q directly (no remapping), since it was derived with $\theta_i = q_i$.
- **Dynamic Parameter Realignment:** CoM coordinates in the URDF are expressed in local link frames. Because DH frames differ in orientation, CoM vectors must be rotated to match the dynamics DH convention. Mismatches cause large errors in the computed gravity torque vector and require manual algebraic correction.
- **Unified Parameter Table:** Adopting a single DH parameter table for both kinematics and dynamics — with d_4 always positive ($+0.10915$ m) — and explicitly decoupling the kinematic angle remapping from the dynamic evaluation eliminates ambiguity. Using two separate sets of parameters (one for FK, one for dynamics) is a common source of bugs and numerical errors in parameter identification.

VI. CONCLUSION

This paper presented a unified workflow to build a digital twin dynamic model for the UR5 manipulator starting from its URDF representation. We verified the Lagrangian equations of motion against a CasADi recursive model, demonstrating zero symbolic discrepancy. Finally, we split the experimental data 50/50 and successfully identified base frame mounting offsets and dynamic parameters (masses and CoMs) using a PD tracking gravity compensation mismatch model. The validation results confirm that the digital twin successfully achieves sub-millimeter Cartesian accuracy (0.73 mm), suitable for high-precision robotic operations.

TABLE II
NOMINAL VS. CALIBRATED LINK DYNAMIC PARAMETERS

Link	Nominal Mass (kg)	Calibrated Mass (kg)	Nominal CoM (m)	Calibrated CoM (m)
1	3.7000	3.7000	[0.0000, 0.0892, 0.0000]	[0.0000, 0.0892, 0.0000]
2	8.3930	8.3913	[0.2800, 0.0000, -0.1358]	[0.2580, 0.0124, -0.1358]
3	2.2750	1.9004	[0.2500, 0.0000, -0.0162]	[0.1500, -0.0295, -0.0162]
4	1.2190	1.1797	[0.0000, -0.0162, 0.0000]	[-0.1000, -0.0161, 0.0305]
5	1.2190	1.4628	[0.0000, 0.0000, 0.0000]	[-0.1000, -0.0123, -0.1000]
6	0.1879	0.1503	[0.0000, 0.0000, 0.0000]	[-0.0156, -0.0260, -0.1000]

REFERENCES

- [1] B. Zhou, Y. Jia, S. Zhang, and M. Chen, “Gtpn\gtcn: A transformer-based hybrid compensation framework integrating kinematic grid sampling and dual-modal strategy for industrial robot error compensation,” *Measurement*, p. 121219, 2026.